1️⃣ Introduction to jQuery and Its Features

🔹 What is jQuery?

jQuery is a fast, lightweight JavaScript library designed to simplify:

- DOM manipulation
- Event handling
- Animations
- AJAX calls
- Cross-browser compatibility

It follows the philosophy:

"Write less, do more."

Official site: **jQuery Foundation**

🔹 Why jQuery Was Created?

Before jQuery:

- Different browsers handled JavaScript differently.
- Writing DOM manipulation required long code.
- AJAX was complex.
- Event handling was inconsistent.

jQuery solved:

- Cross-browser issues
- Long DOM syntax
- Repetitive JavaScript code

🔹 How to Add jQuery to Project

Option 1: CDN (Recommended for teaching/demo)

<script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>

Option 2: Download and use locally

🔹 **Basic jQuery Syntax**

$(selector).action();

- $ → jQuery object
- selector → HTML element
- action() → method (hide, show, click, css, etc.)

🔥 **Features of jQuery**

1️⃣ **DOM Traversing & Manipulation**

- Select elements
- Modify content
- Add/remove elements

2️⃣ **Event Handling**

- click
- change
- submit
- hover

3️⃣ **Effects & Animations**

- hide()
- show()
- fadeIn()
- slideDown()

4️⃣ **AJAX Support**

- $.get()
- $.post()
- $.ajax()

5️⃣ Cross-browser Support

Works consistently across Chrome, Firefox, Edge.

2️⃣ Basic Components of jQuery

Let's break jQuery into its core building blocks.

🔹 1. Selector Engine

jQuery uses CSS-style selectors.

```
$("#id")
$(".class")
$("p")
$("div span")
```

Example

```
<p id="demo">Hello</p>

<script>
$("#demo").hide();
</script>
```

🔹 2. Document Ready Function

Ensures DOM is fully loaded before executing code.

```
$(document).ready(function(){
  // code here
```

```
});
```

Modern shorthand:

```
$(function(){
  // code here
});
```

⚠ Important for beginners — prevents null reference errors.

🔹 3. jQuery Methods

Category	Methods
Content	text(), html(), val()
CSS	css(), addClass(), removeClass()
Effects	hide(), show(), toggle()
Events	click(), on()

🔹 4. Chaining

jQuery supports method chaining.

```
$("#box").css("color","red").slideUp(1000).slideDown(1000);
```

Single line → multiple operations.

3️⃣ DOM Manipulation in jQuery

Now we go deeper technically.

🔹 Changing Content

HTML

```
$("#para").html("<b>Updated</b>");
```

Text

```
$("#para").text("Updated");
```

Difference:

- html() renders HTML
- text() treats as plain text

🔹 Changing CSS

```
$("#box").css("background","blue");
```

Multiple CSS:

```
$("#box").css({
  "color":"white",
  "font-size":"20px"
});
```

🔹 Adding & Removing Classes

```
$("#box").addClass("highlight");
$("#box").removeClass("highlight");
$("#box").toggleClass("highlight");
```

🔹 Creating & Removing Elements

```
$("#list").append("<li>New Item</li>");
$("#item").remove();
```

Other methods:

- prepend()
- before()
- after()

- empty()

4️⃣ Events in jQuery

Events allow user interaction handling.

🔹 Click Event

```
$("#btn").click(function(){
  alert("Button clicked");
});
```

🔹 Using .on() (Best Practice)

```
$("#btn").on("click", function(){
  alert("Clicked");
});
```

Why .on()?

- Supports dynamic elements
- More flexible

🔹 Multiple Events

```
$("#box").on({
  mouseenter: function(){
    $(this).css("background","yellow");
  },
  mouseleave: function(){
    $(this).css("background","white");
  }
```

```
});
```

⚠ When to Use jQuery Today?

Since you are also learning:

- ES6
- Modern JavaScript
- DOM APIs
- Fetch API

Understand this:

Modern JS can do almost everything jQuery does.

jQuery is still useful in:

- Legacy enterprise applications
- ASP.NET MVC older projects
- Quick prototypes